

* NLP algorithm -- Word embedding algorithm

BOW (Bag of Words)

TFIDF (Term Frequency * Inverse document frequency)

WORD2VEC

① BOW (Bag of Words)

Text data

Bag of words

	cat	cute	dog	small
'small dog',	0	0	1	1
'cute cute cat',	1	2	0	0
'cute dog'	0	1	1	0

Q What is the difference b/w one hot encoder vs bow?

In one hot encoder we have binary form only 0 or 1
In BOW we can have 2, 3, 4 etc also.

② $TFIDF = TF * IDF$

~~Sentence~~ $TF = \frac{\text{No. of repetition of words in sentence}}{\text{No. of words in sentence}}$

$IDF = \log \left(\frac{\text{No. of sentences}}{\text{No. of sentences containing words}} \right)$

Sent 1 $\rightarrow$ good boy
 Sent 2 $\rightarrow$ good girl
 Sent 3 $\rightarrow$ boy girl good

Words	Frequency
good	3
boy	2
girl	2

	TF				IDF
	Sent 1	Sent 2	Sent 3	Words	
good	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{3}$	good	$\log(\frac{3}{3}) = 0$
boy	$\frac{1}{2}$	0	$\frac{1}{3}$	boy	$\log(\frac{3}{2})$
girl	0	$\frac{1}{2}$	$\frac{1}{3}$	girl	$\log(\frac{3}{2})$

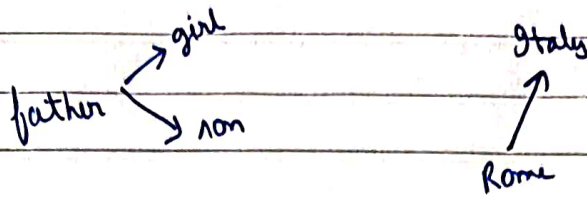
TF * IDF

	good	boy	girl
Sent 1	0	$\frac{1}{2} \log \frac{3}{2}$	0
Sent 2	0	0	$\frac{1}{2} \log \frac{3}{2}$
Sent 3	0	$\frac{1}{3} \log \frac{3}{2}$	$\frac{1}{3} \log \frac{3}{2}$

(document frequency for word) $\log \frac{N}{df}$

③ Word2Vec

Word to Vector



We are trying to find out semantic similarity taken from the paragraph using help of word2vec concept.

word2vec will be slow on big data